EE4-14
EE9-SO16
EE9-AML3-01

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE/EIE PART IV: MEng and ACGI

SPEECH PROCESSING

Thursday, 13 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : W. Dai

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Speech Processing

1. a) i) Describe the process of closed-phase LPC analysis. Include appropriate diagrams and relevant equations. [2]

ii) Explain the process of pre-emphasis in the context of LPC analysis. State the main spectral characteristics of signals in the source-filter speech production model, and the effect of pre-emphasis on these signals. [2]

iii) Explain how pre-emphasis should be used in closed-phase LPC analysis. Include justifications for your explanation. [2]

b) Consider a speech coder employing Code Excited Linear Prediction. The coder uses 10th order autocorrelation LPC analysis.

i) Define Line Spectral Frequencies (LSFs) including appropriate formulae for calculation. [2]

ii) State the advantages of using LSFs in the speech coder under consideration. [2]

iii) Give 3 reasons that bandwidth expansion is used in coders of this type. [2]

iv) Design a bandwidth expansion scheme such that a pole originally at radius 0.994 has a bandwidth after expansion of 20 Hz. Assume the coder operates with a sampling frequency of 8 kHz. [4]

c) Consider a differential speech coding scheme in which $\hat{x}(n)$ represents the signal after quantization such that

$$\hat{x}(n-1) = x(n-1) + e(n-1)$$

where $e(n)$ represents the quantization noise, and the difference signal $d(n)$ is formed as

$$d(n) = x(n) - a\hat{x}(n-1)$$

with constant coefficient a. Show that the variance of $d(n)$ can be written in the form

$$\sigma_d^2 = P\sigma_x^2 + Q\sigma_e^2$$

and find expressions for P and Q.

You may use

$$\rho(i) = \frac{\phi(i)}{\sigma_x^2}$$

where $\phi(i)$ is the autocorrelation function of $x(n)$ and $\rho(i)$ is the corresponding normalized autocorrelation.

[4]

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2. a) Consider a simplified speech production model with the vocal tract filter represented as

$$V(z) = \frac{1 - 2e^{-aT}\cos(bT) + e^{-2aT}}{1 - 2e^{-aT}\cos(bT)z^{-1} + e^{-2aT}z^{-2}}$$

for constants a, b and T.

i) Find the zeros and poles of $V(z)$ and plot the zeros and poles on a labelled diagram of the z-plane. [2]

ii) Find an expression for the impulse response of the simplified vocal tract filter $V(z)$ in terms of $\sin(bT(n+1))$. Consider using partial fraction expansion and the standard z-transform

$$\alpha^n u(n) \xleftrightarrow{\text{z}} \frac{1}{1-\alpha z^{-1}} \quad |z| > |\alpha|.$$

[4]

iii) Draw a labelled sketch of the impulse response for the case $T = 10^{-4}$, $b = 1000\pi$, $a = 200\pi$ and give the amplitude values of the impulse response at $n = 0$, $n = 3$ and $n = 12$. [2]

iv) Draw a labelled sketch of the magnitude response of the vocal tract filter in dB, using the parameters from Part iii). [2]

b) Consider a linear prediction filter of the form

$$A(z) = 1 - \sum_{k=1}^{p} a_k z^{-k}$$

for the case of $p = 3$.

i) Find expressions for the cepstral coefficients, c_k, in terms of the prediction coefficients a_k. [2]

ii) Find an expression in terms of the prediction coefficients a_k for the gain in dB of the filter $V(z) = \frac{1}{A(z)}$ at a normalized frequency of $\pi/3$. [5]

iii) Consider that the prediction coefficients a_k are to be quantized in a speech codec. Using your expression for the gain from your solutions to Part ii), or otherwise, discuss how you would design the quantization scheme for the prediction coefficients for best encoding at a normalized frequency of $\pi/3$, given that the bit-rate is limited. [3]

3. a) State and explain the advantages and disadvantages of applying single-channel speech enhancement to speech signals in

i) telephony applications,

ii) speech recognition applications. [6]

b) Consider a 'smart home' audio device used for speech input. The device has two microphones separated by 20 cm. The signals at the two microphones are

$$x_1(n) = s(n) + e_1(n)$$
$$x_2(n) = s(n-\delta) + e_2(n)$$

where $s(n)$ is the source speech signal at sample index n, $e_1(n)$ and $e_2(n)$ are independent and zero mean noise signals, and δ describes a time-shift (expressed in samples) required to time-align the component $s(n)$ in both microphone signals.

i) Assuming the talker is at the same height as the microphones and at 2.5 m distance, determine the angle of the talker measured relative to the plane of the microphones for the case $\delta = 472\ \mu$s. Use a sketch to clarify your answer and any assumptions you make. Assume that sound waves propagate at 343 m/s. [4]

ii) The two microphone signals $x_1(n)$ and $x_2(n)$ are summed to give

$$y(n) = ax_1(n) + (1-a)x_2(n)$$

where a is a chosen real constant.

Find an expression for the variance of the noise contained in the signal $y(n)$. Assume that δ is known. State any assumptions and clearly define any symbols used. [3]

iii) For the same signal $y(n)$, find an expression for the value of a that minimizes the variance of the noise in $y(n)$. [4]

c) Explain what is meant by the term 'musical noise' in the context of speech enhancement. How can musical noise be minimized in speech enhancement algorithms? Include any relevant mathematical expressions. [3]

4. a) Consider the following utterances spoken by a female speaker and contaminated by babble noise such that the SNR is -1 dB.

Utterance	Spoken Text
A	"It's a pity to recognize speech badly."
B	"Oh two oh, seven five nine four, six two three five."
C	"Hi Jim, it's Mum. Why don't you answer your phone? Call you later."

i) Write down the rank order of the utterances according to the expected recognition accuracy for a typical automatic speech recognition system. [2]

ii) Explain your choice of rank order. [2]

iii) How would the recognition accuracy, and the rank order you have written down, change if the speech signal were to contain significant reverberation in addition to the babble noise? [2]

iv) Rank order the utterances as in Part i) but now for a human listener instead of an automatic speech recognition system, and explain your choice. [2]

b) Compare and contrast the follow approaches to automatic speech recognition:

- automatic speech recognition systems based on neural networks,
- automatic speech recognition systems based on hidden Markov models.

Include in your answer descriptions of the key elements of each approach. [4]

c) Consider a speech recognition system using a hidden Markov model with S states in which the feature vectors for T frames are denoted $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T\}$. The term $d_s(\mathbf{x}_t)$ denotes the probability density of generating an observation frame $\mathbf{x}_t$ from state s. The term $a_{i,j}$ denotes the transition probability from state i to state j of the model.

For a particular example with $T = 6$ and $S = 4$, Table 1 shows the values of $d_s(\mathbf{x}_t)$ and Table 2 shows the state transition probabilities. The symbol '=' indicates the model exit state.

	$d_s(\mathbf{x}_t)$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$	$\mathbf{x}_5$	$\mathbf{x}_6$
	1	0.5	0.4	0.5	0.2	0.1	0.1
State	2	0.4	0.5	0.7	0.6	0.3	0.8
	3	0.2	0.8	0.2	0.8	0.2	0.2
	4	0.5	0.4	0.5	0.2	0.5	0.8

Table 1: Probability Densities

		j				
	$a_{i,j}$	1	2	3	4	=
i	1	0.9	0.1	0	0	0
	2	0	0.5	0.5	0	0
	3	0	0	0.2	0.8	0
	4	0	0	0	0.5	0.5

Table 2: State Transition Probabilities

i) State any structural features of this hidden Markov model.
[1]

ii) Showing all your working, find the total probability of all alignments of the observation feature vectors with the model for which frame 3 is in state 2. Give your answer to 4 significant figures.
[7]